SOLUTION DESIGN · EXECUTIVE BRIEF

Meridian Retail Group — Global Omnichannel Lakehouse

Executive brief: the problem, the value, the architecture in one page, the plan, the cost and how success is measured.

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> **Illustrative — fictional client; all figures illustrative.** Every number in this document (revenue, volumes, timelines, benefits, effort and cost) exists to make the reasoning concrete and checkable. Nothing here is a quotation, a commitment, or a statement about any real organisation's estate.

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1. Executive summary

1.1 The business problem, in five bullets

1. **Meridian cannot see its own customer.** A loyalty member who buys in a Lisbon store, browses on the app in Madrid and returns an item through the German call centre is three unrelated records. 40 million loyalty members are managed as 9 disconnected regional lists.
2. **Stock is in the wrong places.** Replenishment runs on a weekly, country-level forecast produced from data that is between 24 hours and 6 days old. On-shelf availability sits around 94.1% against a 97.0% sector benchmark; the gap is worth an estimated $160–265m of lost sales a year.
3. **The close takes nine days and the answer is contested.** Finance, Commercial and Supply Chain each reconcile from a different extract. Month-end consumes roughly 14 FTE of manual reconciliation and the Board sees trading numbers when they are already history.
4. **The platform is out of road.** The central warehouse runs at 91% of capacity, the overnight batch frequently lands after the 07:00 business deadline, and the analytics cluster is two major versions behind support. Growth in e-commerce and own-brand manufacturing has nowhere to go.
5. **The data team is firefighting, not building.** Approximately 70% of the 48-person data engineering capacity is consumed by incident response, re-runs and manual reconciliation. The BI backlog stands at 340 requests with a median age of 11 months.

1.2 The proposed outcome

A single governed data platform — a **lakehouse** — that holds Meridian's trading, customer, supply chain and finance data once, in one shape, with one set of access rules, and serves it to every consumer: dashboards, forecasting models, the replenishment engine, the app, suppliers and auditors.

The programme replaces the central warehouse and the ageing analytics cluster over 24 months using a **strangler** approach — new consumers are pointed at the new platform first, the old systems are drained rather than lifted, and each decommission is gated on a proven reconciliation. Nothing is switched off on faith.

1.3 Headline benefits — order of magnitude

These are **illustrative ranges**, expressed as annual run-rate at steady state (end of Year 3), with the assumption that drives each one. §4 carries the full working. They are a case for doing the work, not a business plan: each must be re-based on Meridian's own actuals in Phase 0.

Value driver	Mechanism	Annual run-rate (illustrative)	Confidence
Inventory reduction	Daily SKU/store forecast replaces weekly country forecast; safety stock re-cut	$50–75m P&L (on $270–410m working capital release)	Medium-high
On-shelf availability	Real-time inventory position and exception-driven replenishment	$45–75m gross margin	Medium
Markdown optimisation	Price and markdown decisions on daily elasticity rather than calendar rules	$55–110m gross margin	Medium
Personalisation and loyalty	Unified customer view drives offer targeting across 9 e-commerce markets	$25–45m gross margin	Medium-low
Supplier performance recovery	OTIF and quality claims evidenced from shared, agreed data	$15–30m	Medium-high
Waste and shrink (cold chain)	Sensor-driven intervention before an excursion becomes a write-off	$10–20m	Medium
Finance close and reporting effort	9-day close to 4-day close; reconciliation automated	$6–9m cost, plus decision speed	High
Legacy run-cost avoidance	Warehouse and cluster decommission; licence and hardware exit	$9–14m	High
Indicative total		**$215–378m**	

Programme investment (illustrative): $28–42m across 24 months, including services, platform consumption, dual-run and client-side backfill. Payback on the *credible lower half* of the benefit range falls inside Year 2. The logic is in §10.6.

1.4 Timeline

Phase	Weeks	What the business gets
0 — Mobilise and prove	1–6	Foundations live; access, security and network resolved; benefits re-based on actuals
1 — Thin slice	7–16	**First value at week 10**: daily store-level sales and stock position for one region, with a governed dashboard replacing four spreadsheets
2 — Core domains	17–40	Customer 360, unified sales, inventory and product; forecasting in pilot
3 — Scale and migrate	41–72	All 14 countries; warehouse consumers migrated; dual-run and reconciliation
4 — Decommission and hand over	73–96	Legacy warehouse and cluster off; client team running the platform

1.5 The ask

#	Decision requested	By
1	Approve Phase 0 and Phase 1 (weeks 1–16) at an illustrative **$3.4–4.3m** of services plus platform consumption, with a formal gate at week 16 before Phase 2 is committed.	ExCo
2	Name a single accountable **executive sponsor** with authority across Commercial, Supply Chain, Finance and IT. A steering committee is not a sponsor.	CEO
3	Mandate **SAP data access** for the programme, including the basis team's time. This is the single most common cause of delay in this class of programme (§11, R2).	CFO / CIO
4	Start the **network and connectivity work in week 1** (private connectivity, firewall rules, identity federation). Lead times here, not engineering, set the critical path (§11, R8).	CIO
5	Accept the **decommission gates in §8.6** as binding — including the commitment that the legacy warehouse is switched off, not left running "just in case". The benefit case depends on it.	CFO

2. Current state

2.1 The estate

Layer	System	Scale	Condition
Central warehouse	On-premises MPP appliance (Teradata-class)	480 TB, ~11,000 tables, ~2,900 registered reports	91% capacity; next hardware refresh due in 14 months; support renewal in 9
Analytics cluster	Hadoop (CDH-lineage), 120 nodes, HDFS + Hive + Spark 2.x	2.1 PB raw, ~60% cold	Two major versions behind support; no clear upgrade path; 3 of 5 original engineers have left
ETL	Legacy graphical ETL tool	43 point-to-point job streams, ~2,700 individual jobs	No version control on 60% of jobs; change lead time 6–8 weeks
ERP	SAP ECC 6.0, single global instance, 3 company-code groups	~8,500 active tables; finance, inventory, procurement, own-brand production	Stable; heavily customised; the extraction conversation is political
CRM	Salesforce (Sales + Service Cloud) — B2B wholesale and franchise	~4.2m records	Well run; API limits are the constraint

Layer	System	Scale	Condition
POS	Three regional POS platforms (EU, APAC, Americas)	2,400 stores	Different schemas, different close-of-day timing, different offline behaviour
Digital	Homegrown clickstream pipeline, Kafka to HDFS	~360m events/day	Built by a team that no longer exists; no schema registry discipline
Last mile	Excel and Google Sheets	~1,400 "critical" spreadsheets identified in a 2025 audit	This is where most decisions are actually made

2.2 The pain, in business terms

Month-end close takes nine working days. Days 1–4 are spent waiting for the overnight batch to produce a clean trial position; days 5–7 are reconciliation between the warehouse, SAP and the regional POS totals; days 8–9 are commentary. Roughly 14 FTE across Group and regional finance are consumed each month, and the Board's trading review discusses a month that closed nine days ago.

Stock-outs are structural, not incidental. Replenishment proposals are generated weekly from a country-level forecast. A store that sells out on a Tuesday is not corrected until the following Monday's run. Measured on-shelf availability is 94.1%; the top-500 SKUs sit at 96.3% and the long tail at 91.8%. Commercial estimates — and the programme should validate — that closing half the gap to the 97.0% benchmark is worth 0.6–1.0% of store revenue.

There is no single customer view. Loyalty is administered per country. A member's e-commerce identity is an email address; their store identity is a card number; their service identity is a Salesforce contact. Match rates between the three are unknown because nobody has ever computed them. Marketing spends an estimated $18–24m a year on audiences it cannot deduplicate.

The data team spends 70% of its time firefighting. The 48-person team logged 1,190 production incidents in the last 12 months. The top three causes are upstream schema changes arriving without notice, late-arriving POS files from stores with poor connectivity, and batch windows overrunning into business hours. None of these are hard problems; they are unmanaged ones.

Everything important ends in a spreadsheet. The trading pack, the markdown decision, the supplier scorecard and the availability review are all assembled by hand. The numbers in them do not always agree, and no two are assembled the same way twice.

2.4 What is genuinely good, and must be preserved

Not everything must be replaced. The programme explicitly protects three things:

- **The SAP finance chart of accounts and the company-code hierarchy.** These are correct, governed and auditable. The lakehouse conforms to them; it does not reinvent them.
- **The Salesforce B2B process.** The wholesale team's way of working is effective. The platform reads Salesforce; it does not attempt to re-platform it.
- **The store-level merchandising hierarchy** maintained in the PIM. It is imperfect, but it is the one dimension the whole business already agrees on. It becomes the conformed product dimension.

4. Use cases and value

Eight use cases are carried into the plan. They were selected because each one is (a) owned by a named executive, (b) measurable against an existing KPI, and © dependent on data the programme is already ingesting for another reason. Use cases that fail any of those three tests are parked, not refused.

A note on latency. The column "freshness it truly needs" is the most commonly over-specified number in a programme like this. Every stakeholder asks for real time. Only two of these eight use cases genuinely require it; asking for it elsewhere converts a cheap batch pipeline into an expensive streaming one for no decision benefit. The figures below are the ones the programme will defend.

UC-1 — Unified Customer 360

Business question	Who is this customer across every channel and country, what have they bought, what have they returned, what have they been sent, and what have they consented to?
Consumers	Marketing (campaign and audience build), Customer Service, Loyalty, Digital product teams, Group Commercial
Freshness truly needed	**Daily** for the analytical profile; **within 1 hour** for consent and suppression state; **near-real-time** only for the subset of attributes used by on-site personalisation (UC-6)
Depends on	POS (3.3), clickstream (3.4), Salesforce (3.2), marketing platforms and consent store (3.8), loyalty master (within POS/SAP), reference (3.12)
Value driver	Removes duplicate audience spend; makes every other customer-facing use case possible. **It is the enabling asset, and its value is mostly realised through UC-4 and UC-6.**
Value range	$8–14m direct (deduplicated media spend and service efficiency), plus it is a hard dependency for $25–45m in UC-6
Key assumption	Identity resolution achieves a 70–80% match rate between store loyalty card and e-commerce account within the loyalty population. If the true rate is below 55%, UC-6's value halves. **Measure this in Phase 0 on a sample — do not assume it.**

UC-2 — Demand forecasting and replenishment

Business question	How many units of each SKU will each store sell tomorrow, and what should be ordered today?
Consumers	Supply Chain planning, the replenishment engine (as an automated consumer), store operations
Freshness truly needed	**Daily, completed by 04:00 local** so the order cut-off is met. Intraday is not required; the order is placed once
Depends on	POS (3.3), SAP inventory and purchasing (3.1), PIM (3.5), EDI ASN (3.6), weather and calendar (3.9), promotional calendar (marketing, 3.8)
Value driver	Forecast error reduction lets safety stock be re-cut. Inventory is ~$3.4bn; carrying cost ~18%
Value range	**$50–75m annual P&L**, on a working-capital release of $270–410m
Key assumption	Forecast accuracy (WMAPE at SKU-store-day) improves from ~38% to 26–30%, and Supply Chain actually re-cuts safety-stock parameters on the back of it. **The second half of that sentence is where these programmes fail: a better forecast that nobody acts on is worth zero.** Make the parameter re-cut an explicit, owned deliverable

UC-3 — Real-time inventory availability

Business question	What is actually on the shelf, in the back room, in the DC and in transit, right now, for this SKU at this store?
Consumers	E-commerce (click-and-collect promise), store colleagues (app), the availability exception process, customer service
Freshness truly needed	**15 minutes.** This is one of the two genuine near-real-time cases. A collect promise made on 24-hour-old stock is a broken promise
Depends on	POS intraday (3.3), SAP goods movements (3.1), EDI ASN (3.6), PIM (3.5)
Value driver	On-shelf availability, plus reduced failed click-and-collect orders (currently ~3.1% of collect orders substitute or cancel)
Value range	**$45–75m gross margin**, from recovering 0.6–1.0% of store revenue on a 28% gross margin, phased over Years 2–3
Key assumption	The stock position derived in the lakehouse is accurate enough to be trusted operationally. It will not be on day one — book stock accuracy is estimated at 88% at SKU-store level. **Phase 2 must include a stock-accuracy measurement and a correction loop, or this use case will be rejected by stores**

UC-4 — Markdown and promotional optimisation

Business question	For this SKU in this store, what price or markdown maximises margin over the remaining life of the stock?
Consumers	Commercial / category managers, Pricing, regional trading teams
Freshness truly needed	**Daily.** Markdown decisions are taken on a weekly cycle with daily exception review
Depends on	POS (3.3), SAP inventory and valuation (3.1), PIM (3.5), competitor pricing and panel data (3.9), UC-2 forecast output
Value driver	Markdown spend is estimated at ~5.6% of store revenue ($1.5bn). Sector experience suggests 3–6% improvement is achievable when decisions move from calendar rules to measured elasticity
Value range	**$55–110m gross margin**
Key assumption	Category managers accept a recommendation engine in the loop. Recommend a **recommend-and-override** design, not automation — adoption is the binding constraint, not model quality

UC-5 — Supplier performance and claims recovery

Business question	Which suppliers are delivering on time and in full, at what quality, and what are we owed?
Consumers	Procurement, Supply Chain, Finance (claims), and **the suppliers themselves**
Freshness truly needed	**Daily**, with a weekly published scorecard
Depends on	EDI (3.6), SAP purchasing and goods receipt (3.1), DC IoT and quality events (3.7), PIM supplier linkage (3.5)
Value driver	OTIF penalties and quality claims that are currently not raised because they cannot be evidenced
Value range	**$15–30m**, of which roughly half is claims recovery and half is behavioural improvement once suppliers see the same scorecard
Key assumption	Suppliers can be given governed read access to their own slice of the scorecard without a per-supplier integration project. This is the case that justifies OpenSharing (§7.6)

UC-6 — Personalisation and offer targeting

Business question	What should we show or offer this customer, now, given what they have bought, browsed and consented to?
Consumers	Digital product, Marketing, the app and website, the loyalty offer engine
Freshness truly needed	**Session-level (seconds) for on-site behavioural signals**; daily for the profile attributes. This is the second genuine real-time case, and only for a narrow feature set
Depends on	UC-1 Customer 360, clickstream (3.4), POS (3.3), PIM (3.5), consent store (3.8)
Value driver	Incremental revenue on loyalty-attributed sales, currently ~44% of store revenue
Value range	**$25–45m gross margin**
Key assumption	Incremental lift of 1.5–3% on targeted segments, proven by holdout testing. **Require a holdout on every campaign from day one.** Without holdouts the value is unprovable and the programme will be argued about rather than extended

UC-7 — Finance close acceleration

Business question	What did we earn, where, and can we prove it — by day 4 rather than day 9?
Consumers	Group Finance, regional controllers, Investor Relations, external audit
Freshness truly needed	**Daily by 06:00 local**, with a flash trading position available intraday
Depends on	SAP finance (3.1), POS (3.3), reference and FX (3.12), legacy EDW history (3.11) during dual-run
Value driver	Reconciliation effort, decision latency, and the audit-evidence burden
Value range	**$6–9m cost**, principally 14 FTE of monthly reconciliation reducing to 4–5, plus a materially cheaper audit cycle. The larger, unquantified benefit is that the Board argues about the business rather than about the numbers
Key assumption	Finance accepts lakehouse-derived figures as the reporting basis after a **3-month parallel run with zero unexplained variance**. Do not promise a shorter parallel

UC-8 — Cold-chain compliance and waste reduction

Business question	Was this product held within temperature specification along its whole journey, and where are we losing product to excursions?
Consumers	Quality and food safety, DC operations, store operations, external auditors and regulators
Freshness truly needed	**1 minute for alerting** (operational, on the streaming path); daily for analysis and evidence
Depends on	IoT (3.7), EDI ASN (3.6), SAP movements (3.1), PIM product attributes (3.5)
Value driver	Product written off after an excursion, plus the cost of manual compliance evidence gathering
Value range	**$10–20m**, plus a materially reduced regulatory risk that is not monetised here
Key assumption	Alerts reach a person who can act within the product's thermal tolerance. **An alert with no owner is a log line.** The operating-model change is bigger than the engineering

4.9 Prioritisation matrix

Scores are 1–5. **Value** is annual run-rate benefit. **Effort** is delivery effort (5 = least effort). **Data readiness** is how usable the required sources are today (5 = ready). **Priority score** = Value x (Effort + Readiness) / 2, rounded. This is a conversation starter for the sponsor, not an algorithm; the sequencing in §8 also reflects dependency order.

Use case	Value	Effort (5 = easy)	Data readiness	Priority score	Phase
UC-7 Finance close	2	4	4	8	1–2
UC-3 Real-time availability	5	2	3	13	2–3
UC-2 Demand forecasting	5	2	3	13	2–3
UC-4 Markdown optimisation	5	3	3	15	3
UC-5 Supplier performance	3	4	3	11	2
UC-1 Customer 360	3	3	2	8	2
UC-6 Personalisation	4	2	2	8	3–4
UC-8 Cold chain	2	3	4	7	3

How to read this. UC-4 scores highest but depends on UC-2's forecast, so it cannot go first. UC-7 scores modestly but is sequenced early because it is the **credibility purchase**: a finance close that lands on day 4 is the outcome that buys the programme its Phase 3 budget. UC-1 scores lower than its strategic importance because its data readiness is genuinely poor — that is a finding, not a reason to skip it.

Deliberately excluded from Phase 1–3 scope, with the reason:

Parked use case	Why not now
Store labour optimisation	Depends on worker-level HR data, which is blocked by works-council consultation (§3.10)
Assortment and space planning	Requires planogram data that is not currently captured digitally in 6 of 14 countries
Dynamic pricing (as opposed to markdown)	Regulatory and brand risk in grocery; a separate decision, not a data decision
Scope-3 emissions reporting	Real and coming, but the data standard is still moving; revisit at the Phase 3 gate
Generative assistants over the estate	Becomes cheap and safe **after** the governed semantic layer exists, not before. Sequence it into Phase 4, not Phase 1

5. Target architecture

5.1 The shape, in one paragraph

Every source lands **once**, in its original grain, into a **bronze** zone owned by the platform team. Domain teams transform bronze into **silver** — conformed, deduplicated, quality-gated business entities with stable keys. **Gold** holds the consumption-shaped products: dimensional models, metric definitions, feature tables and extracts. One catalog per environment, one schema per domain, one owner per schema. Consumption is through governed SQL,

dashboards, a natural-language interface and model serving. Nothing reads a source system directly except an ingestion pipeline, and nothing writes to gold except a pipeline in version control.

5.8 Architecture decisions, consolidated

ID	Decision	Rejected alternative
D-01	Ladder-first ingestion: most managed mechanism that meets the need	One uniform pattern for all sources
D-02	Declarative pipelines as the default transformation framework, orchestrated by Lakeflow Jobs	Hand-written notebook pipelines
D-03	SAP via log-based CDC, ODP/SLT held as a costed fallback	Nightly full export; SAP-native extraction as primary
D-04	Scheduled available-now triggers unless a use case demands continuous	Always-on streaming everywhere
D-05	Catalog per environment, schema per domain	Catalog per domain; single catalog with prefixes
D-06	ABAC policies on governed tags for masking and row filtering	Per-table masks and filters as the primary mechanism
D-07	Serverless SQL warehouses by default	Provisioned classic warehouses sized to peak
D-08	Three regional production workspaces, regional metastores, cross-region via sharing	Single global workspace; workspace per country
D-09	Cost attribution built in Phase 0	Cost management deferred to Phase 3
D-10	Infrastructure and workloads deployed as **Declarative Automation Bundles** from Git (§9.4)	UI-created jobs promoted by export/import
D-11	Data products published as governed **schemas with contracts** (§6)	A central team owning all transformation
D-12	Legacy EDW read by federation during migration; only unrecoverable history is copied	Full warehouse migration
D-13	Pseudonymisation at the bronze-to-silver boundary (§7.2)	Full-estate masking policies over identifier-bearing tables
D-14	Supplier exchange via OpenSharing (§7.6)	A supplier portal; 1,200 per-supplier extracts

D-10 is stated in full in §9.4, D-11 in §6.1, D-13 in §7.2 and D-14 in §7.6, alongside the material they govern.

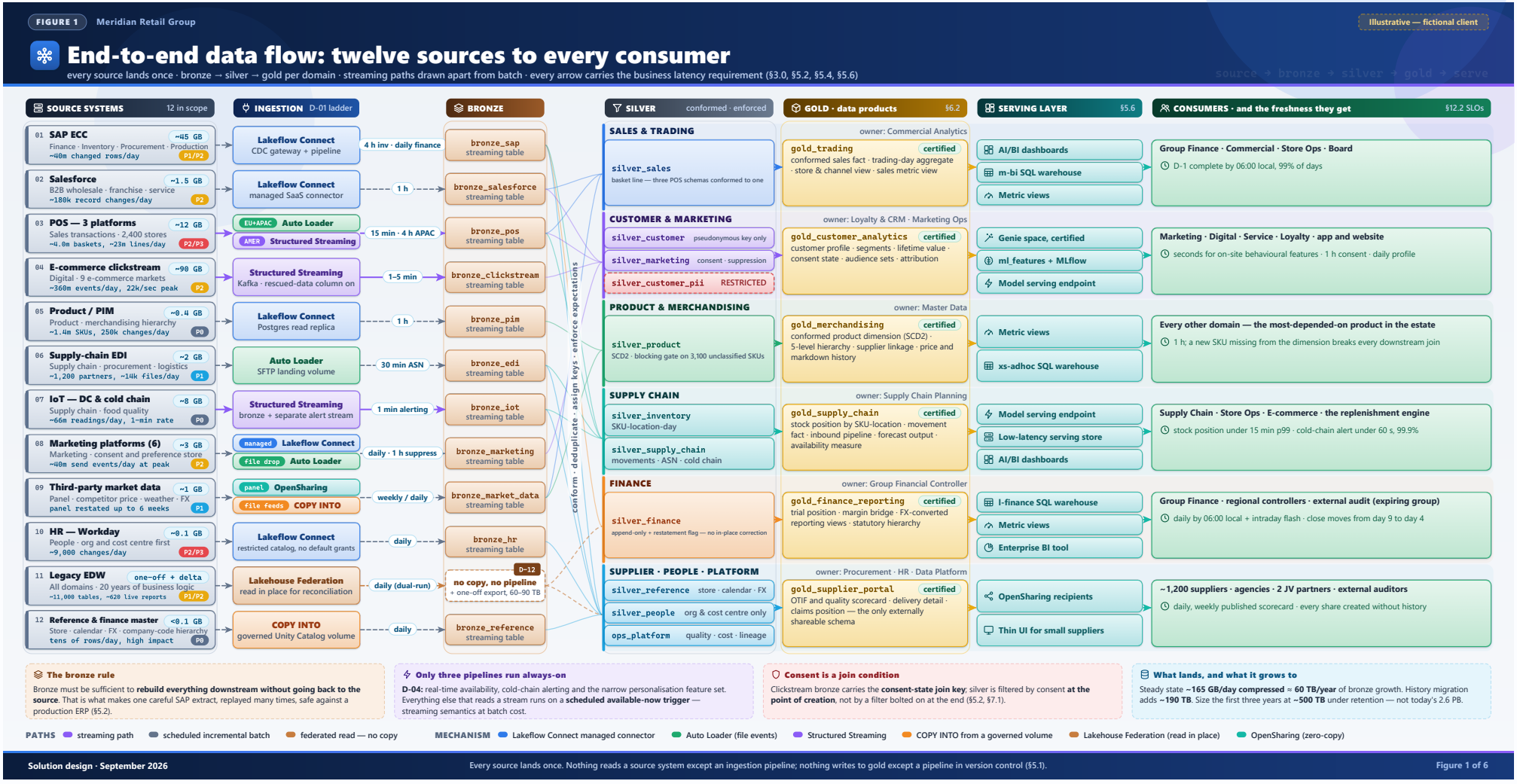


Figure 1 — End-to-end flow

`case01_fig1_end_to_end_flow.svg`

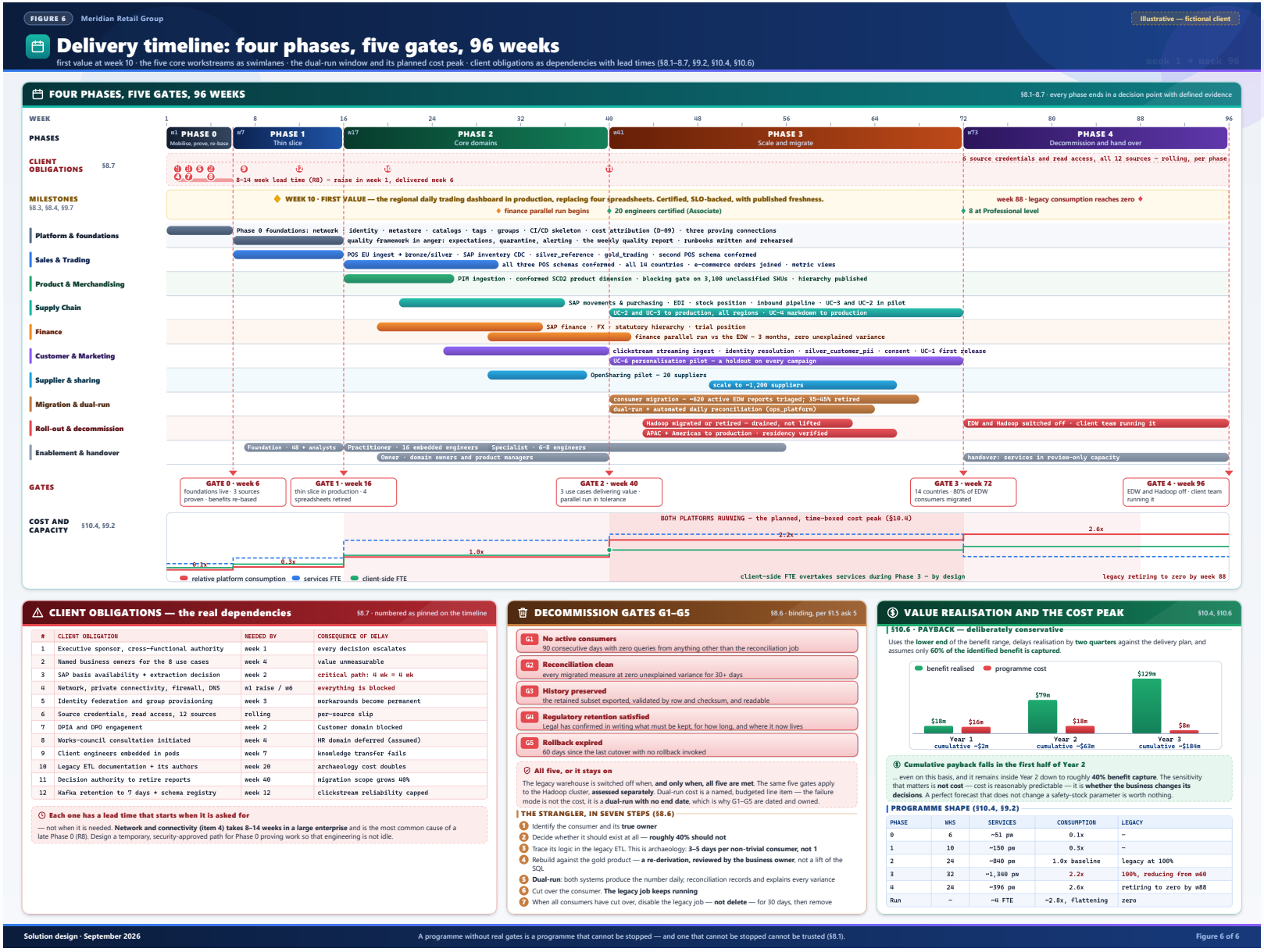


Figure 6 — Delivery timeline

`case01_fig6_delivery_timeline.svg`

8. Delivery plan

8.1 Shape of the programme

Four phases over 96 weeks. Each phase ends in a **gate**: a decision point with defined evidence, at which the programme can be stopped, re-scoped or continued. A programme without real gates is a programme that cannot be stopped, and one that cannot be stopped cannot be trusted.

Phase	Weeks	Theme	Gate evidence
0	1–6	Mobilise, prove, re-base	Platform foundations live; three source connections proven end to end; benefits re-based on client actuals; Phase 1 plan confirmed
1	7–16	Thin slice	One region, one domain, in production, with a real consumer using it in anger; decommission of at least 4 spreadsheets
2	17–40	Core domains	Customer, Product, Sales, Supply Chain and Finance silver and gold live for all regions; forecasting in pilot; 3 use cases delivering measured value
3	41–72	Scale and migrate	All 14 countries; 80% of active EDW consumers migrated; dual-run reconciliation passing; Hadoop workloads migrated or retired
4	73–96	Decommission and hand over	EDW and Hadoop off; client team running the platform; services role reduced to advisory

8.2 Phase 0 — Mobilise and prove (weeks 1–6)

Week	Deliverable
1	Cloud accounts, network design signed off, **private connectivity requests raised** (the long pole — start on day one). Identity federation design agreed
1–2	Landing-zone triage: what in the Hadoop cluster is worth keeping. Expect to recommend leaving 60%+ behind
2	Unity Catalog metastore, catalogs, initial schemas, group taxonomy, governed tags defined
2–3	CI/CD skeleton: repository structure, bundle definitions, dev/test/prod targets, first automated deployment
3–4	Three proving connections: SAP (one table group), POS EU (one country), PIM. Each landing into bronze, end to end
3–4	Cost attribution reporting live (D-09)
4–5	Benefits re-based: identity match-rate sample for UC-1, forecast-accuracy baseline for UC-2, availability measurement basis for UC-3
5	PII discovery pass over the in-scope EDW subset and the three proven sources
5–6	Phase 1 backlog agreed with named business owners; gate pack prepared

What "proven" means: data lands, is queryable, has a quality expectation attached, has lineage, is tagged, appears in cost reporting, and was deployed by the pipeline rather than by hand. Anything less is a demonstration, not a proof.

8.3 Phase 1 — The thin slice (weeks 7–16), first value at week 10

The thin slice is **one region (EU), one business outcome (daily trading and stock position), one consumer group (regional Commercial and Supply Chain)**. It is deliberately narrow and deliberately vertical: it exercises every layer of the architecture rather than building one layer broadly.

Week	Deliverable
7–8	POS EU ingestion at full country scope, bronze to silver conforming for one of three POS schemas
7–9	SAP inventory movements CDC into bronze and silver
8–9	`silver_reference` (store master, calendar, FX) with a named owner and a maintenance process
9	`gold_trading` first product: daily sales by store, SKU and day, with margin
10	**First value: the regional daily trading dashboard in production**, replacing four spreadsheets. Certified, SLO-backed, with published freshness
10–12	Stock position product; availability measure defined and agreed (expect a fortnight of argument about the definition — budget for it, it is real work)
12–14	Quality framework in anger: expectations, quarantine, alerting, the weekly quality report
14–15	Second POS schema conformed (proves the pattern generalises)
15–16	Gate pack: measured freshness, measured quality, measured cost per product, user adoption, and a candid list of what was harder than expected

Why week 10 and not week 6. A dashboard can be produced in three weeks. A dashboard that is governed, tested, SLO-backed, cost-attributed, deployed from version control and owned by a named business team takes ten. The second one takes two. Phase 1 buys the *second* one.

8.4 Phase 2 — Core domains (weeks 17–40)

Five workstreams run in parallel, each with its own domain owner:

Workstream	Weeks	Lands
Sales & Trading	17–30	All three POS schemas conformed; all 14 countries; e-commerce orders joined; `gold_trading` complete with metric views
Product & Merchandising	17–26	PIM ingestion; conformed SCD2 product dimension; the "to be classified" quality gate; hierarchy published
Supply Chain	22–36	SAP movements and purchasing; EDI; stock position; inbound pipeline; UC-3 in pilot; UC-2 forecast in pilot
Finance	20–34	SAP finance; FX; statutory hierarchy; trial position; **parallel run against the EDW begins at week 30**
Customer	26–40	Clickstream streaming ingest; identity resolution; `silver_customer_pii` and pseudonymisation; consent integration; UC-1 first release

Cross-cutting in the same window: supplier scorecard and the first OpenSharing pilot with 20 suppliers (weeks 30–38); analyst onboarding at scale from week 24; the enablement programme (§9.6) running continuously from week 17.

Gate at week 40: three use cases delivering measured value; finance parallel run showing variance inside tolerance; at least two domains operating with genuine client-side ownership.

8.5 Phase 3 — Scale and migrate (weeks 41–72)

This is where the programme either succeeds or becomes permanent. Discipline is everything.

Workstream	Weeks	Lands
Consumer migration	41–68	The ~620 active EDW reports triaged: migrate, rebuild, merge or retire. Expect **35–45% to be retired outright** — that is the normal finding and it must be presented as a win
Dual-run and reconciliation	41–64	Automated daily reconciliation between EDW and lakehouse for every migrated measure
Hadoop migration	44–62	Workloads migrated or retired; retained history moved; the cluster drained rather than lifted
Use case scale-out	41–72	UC-2 and UC-3 to production across all regions; UC-4 markdown to production; UC-6 personalisation pilot
Regional roll-out	44–66	APAC and Americas workspaces to full production; residency controls verified per country
Supplier sharing	50–66	Scale from 20 to 1,200 suppliers

8.6 Migration approach — strangler, dual-run, decommission gates

The strangler pattern, applied concretely. The EDW is not migrated. Its *consumers* are migrated, one at a time, to products built from source. Each migration follows the same seven steps:

1. Identify the consumer (a report, an extract, a downstream system) and its true owner.
2. Determine whether it should exist at all. Roughly 40% should not.
3. Trace its business logic in the legacy ETL and document it. This is archaeology, and it is the expensive part — budget 3–5 days per non-trivial consumer, not 1.
4. Rebuild the logic against the lakehouse gold product. Not a lift of the SQL — a re-derivation, reviewed by the business owner.
5. **Dual-run** for a defined period: both systems produce the number daily, and an automated reconciliation job compares them, records the variance, and explains every difference above tolerance.
6. Cut over the consumer. The legacy job keeps running.
7. When all consumers of a legacy job have cut over, the legacy job is disabled — not deleted — for 30 days, then removed.

Reconciliation is a product, not an activity. Build it in Phase 2, before it is needed: an automated job, a reconciliation results table in `ops_platform`, a dashboard showing every migrated measure's variance over time, and a rule that a measure must show **zero unexplained variance for 30 consecutive days** before its consumer may cut over. "Explained" means written down, agreed and attributed to a specific cause — usually a defect in the legacy logic, which is itself a valuable finding to publish.

Decommission gates. The legacy warehouse is switched off when, and only when, all five are met:

Gate	Evidence
G1 — No active consumers	90 consecutive days with zero queries from anything other than the reconciliation job
G2 — Reconciliation clean	Every migrated measure at zero unexplained variance for 30+ days
G3 — History preserved	The retained history subset exported, validated by row and checksum, and readable
G4 — Regulatory retention satisfied	Legal has confirmed, in writing, what must be kept and for how long, and where it now lives
G5 — Rollback expired	60 days since the last cutover with no rollback invoked

The same five gates apply to the Hadoop cluster, assessed separately.

Dual-run cost is a named, budgeted line item, not an overrun. Running both platforms for 24 weeks is expensive and entirely predictable; §10.4 carries it explicitly. The failure mode here is not the cost — it is a dual-run with no end date, which is why G1–G5 are dated and owned.

8.7 What the client must provide

The programme cannot deliver these, and each one has a lead time that starts when it is asked for, not when it is needed.

#	Client obligation	Needed by	Consequence of delay
1	Executive sponsor with cross-functional authority	Week 1	Prioritisation stalls; every decision escalates
2	Named business owners for each of the 8 use cases	Week 4	Value is unmeasurable and unattributable
3	SAP basis team availability and an extraction decision	Week 2	Critical path; a 4-week delay here is a 4-week delay to Phase 2
4	Network and private connectivity, firewall and DNS	Week 1 request, week 6 delivery	Everything is blocked. This is the most common cause of a late Phase 0
5	Identity federation and group provisioning	Week 3	Access model cannot be implemented; workarounds become permanent
6	Source system credentials and read access, all 12 sources	Rolling, per phase	Per-source slip
7	Data protection impact assessment and DPO engagement	Week 2	Customer domain (Phase 2) blocked
8	Works-council consultation initiated	Week 4	HR domain deferred (already assumed)
9	Client-side engineers embedded in pods, from Phase 1	Week 7	Knowledge transfer fails; the programme never ends
10	Legacy ETL documentation and access to the people who wrote it	Week 20	Migration archaeology cost doubles
11	Decision authority to retire reports	Week 40	Migration scope grows by 40%
12	Kafka retention increase to 7 days and a schema-registry commitment	Week 12	Clickstream reliability capped; incidents recur

10. Cost model and commercials

> **No list prices appear in this section.** Databricks and cloud pricing change, vary by region, cloud, commit level and contract, and are the wrong thing to hard-code into a business case. What follows is the **structure** of the cost: the drivers, the ratios we typically observe, and the controls. A real estimate is produced by applying the client's measured workload profile to the current price lists during Phase 0 — and it is produced jointly with the Databricks account team and the cloud provider, because commit structure materially changes the answer.

10.1 The cost drivers

Driver	What moves it	Typical share of run cost
Platform consumption — data engineering (pipelines, jobs, streaming)	Volume ingested, transformation complexity, streaming vs batch, trigger frequency, whether pipelines are always-on	35–50%
Platform consumption — SQL and BI	Number of analysts, dashboard refresh frequency, query efficiency, warehouse auto-stop discipline, how much gold is pre-aggregated	25–40%
Platform consumption — ML (training, features, serving)	Model count, retraining frequency, serving request volume and latency target	5–15%
Cloud storage	Total retained volume, retention policy, file layout, how aggressively maintenance runs	5–10%

Driver	What moves it	Typical share of run cost
Cloud egress and networking	Cross-region replication, cross-cloud sharing, private connectivity	2–8%, and the most commonly forgotten
Ingestion-specific fixed costs	A CDC ingestion gateway runs on classic compute in the client's network and is **billed continuously, including while the ingestion pipeline is idle** — this makes the SAP connector a standing cost, not a per-run one	Material for SAP; budget it explicitly
Migration effort (one-off)	Number of consumers, archaeology depth, reconciliation scope	See §10.4
Dual-run (one-off, time-boxed)	Legacy platform run cost for the overlap period, plus the reconciliation workload	See §10.4

The three levers that move platform consumption most, in our experience, in order:

1. **Trigger frequency and always-on pipelines.** The gap between a continuously running stream and the same pipeline on a scheduled available-now trigger is very large, and for most workloads the business cannot tell the difference. This is D-04, and it is the single biggest lever.
2. **Bytes scanned.** Good file layout (clustering on the real filter columns), automatic table maintenance, and materialising the aggregates that dashboards actually query. A dashboard that scans a year of basket lines on every refresh is the classic and expensive mistake.
3. **Idle compute.** Auto-stop, serverless where demand is bursty, and never letting analysts create their own long-running compute.

10.4 Illustrative effort and cost profile

Effort in person-weeks from §9.2. **Cost columns are illustrative placeholders** for the shape of the profile, not a quotation.

Phase	Weeks	Services person-weeks	Platform consumption (relative)	Legacy dual-run	Notes
0	6	~51	0.1x	—	Small footprint; benchmarking workloads only
1	10	~150	0.3x	—	One region, one domain
2	24	~840	1.0x (baseline)	Legacy continues at 100%	Five parallel workstreams; the peak of services effort
3	32	~1,340	2.2x	Legacy at 100%, reducing from week 60	**Peak total cost.** Both platforms running, migration at full rate
4	24	~396	2.6x	Legacy retiring to zero by week 88	Consumption rises as adoption rises; services fall sharply
Run (Yr 3+)	—	~4 FTE advisory	~2.8x, then flattening	Zero	Steady state

Two things to say plainly to the sponsor:

- **Phase 3 is the expensive one**, because both platforms run at once. This is visible, planned and time-boxed by the gates in §8.6. A programme that hides this is a programme that will have an uncomfortable conversation in month 14.
- **Platform consumption keeps rising after delivery stops**, because success means more people using more data. That is the right outcome, and it is why per-product cost attribution matters more than a total budget number. The metric to manage is **cost per unit of consumption**, not total cost.

10.5 The commercial construct we would propose

Phase	Construct	Rationale
0	Fixed price	Scope is well defined; the client should not carry discovery risk
1	Fixed price against a defined thin slice	We should carry the risk of our own first-delivery estimate
2	Capped time and materials per workstream, with per-product acceptance	Scope genuinely emerges; a fixed price here is priced with a risk premium the client pays regardless
3	Capped time and materials, plus a decommission-linked element	Aligns our incentive with the client's actual goal, which is switching the legacy platform off
4	Reducing retainer, stepped down quarterly against the handover criterion in §9.7	Makes our exit a contractual event rather than a conversation

10.6 Payback logic

Illustrative, and deliberately conservative — it uses the **lower** end of the benefit range, delays benefit realisation by two quarters against the delivery plan, and assumes only 60% of the identified benefit is actually captured.

	Year 1	Year 2	Year 3
Benefit realised (60% of low case, phased)	~$18m	~$79m	~$129m
Programme cost (services + platform + dual-run)	~$16m	~$18m	~$8m
Net	~+$2m	~+$61m	~+$121m
Cumulative	~$2m	~$63m	~$184m

Cumulative payback falls in the first half of Year 2 even on this conservative basis. The sensitivity that matters most is not cost — cost is reasonably predictable — it is **whether the business changes its decisions**. A perfect forecast that does not change a safety-stock parameter is worth nothing. That is why §12 measures business outcomes, not platform outputs, and why §8.7 asks for named business owners in week 4.

12. Success metrics

Two panels. The executive panel measures whether the business changed. The platform panel measures whether the platform is trustworthy. Both are published; the second is the leading indicator of the first.

12.1 Business KPIs

KPI	Baseline	Target (by)	How measured
On-shelf availability	94.1%	96.0% (end Yr 2), 96.8% (end Yr 3)	Existing availability audit process, unchanged, so the comparison is honest
Inventory value	$3.4bn	$3.05–3.15bn at constant sales	SAP inventory valuation, monthly average
Forecast accuracy (WMAPE, SKU-store-day)	~38%	26–30% (end Yr 2)	Held-out backtest on a fixed evaluation set, re-run monthly
Markdown as % of revenue	5.6%	5.2–5.4% at constant sell-through	Finance reporting; sell-through controlled for

KPI	Baseline	Target (by)	How measured
Month-end close	9 working days	4 working days (end Yr 2)	Date of final signed trial position
Finance reconciliation effort	~14 FTE/month	4–5 FTE/month	Time recording, existing process
Customer identity match rate	Unknown	Measured in Phase 0, then 70–80%	Match rate between store loyalty and e-commerce identity within the loyalty population
Loyalty-attributed sales share	44%	48% (end Yr 3)	POS loyalty capture rate
Supplier OTIF	87.4%	92% (end Yr 3)	Scorecard, shared with suppliers, so the number is agreed rather than argued
Cold-chain excursion write-off	Baseline in Phase 2	-30%	Quality system, linked to sensor evidence
Active reports in use	~620 of 2,900	~400 certified products, 2,900 retired	Catalogue and usage telemetry. **Fewer is the target**
Time from question to answer (new analysis)	Median 11 months (backlog age)	Median under 5 working days for self-service; under 6 weeks for a new product	Request tracking

12.2 Platform SLOs

SLO	Baseline	Target	How measured
Certified product freshness	Not measured	Day D-1 complete by 06:00 local, 99% of days	Freshness monitoring on every certified product; breaches logged
Real-time inventory latency	24 h+	Under 15 min, 99th percentile	End-to-end timestamp from POS event to queryable stock position
Cold-chain alert latency	Not measured	Under 60 s, 99.9%	Sensor timestamp to alert dispatch
Pipeline success rate	~89% (est.)	99.5% first-run success	Job run telemetry
Data quality pass rate	Not measured	Over 99.5% of rows passing expectations on certified products; 100% of breaches investigated	Pipeline expectation metrics, published weekly
Incidents per month	~99	Under 15 by end Yr 2	Incident management
Firefighting share of engineering time	~70%	Under 25% by end Yr 2	Time allocation. **This is the metric the engineering team will care about most, and it should be published**
Mean time to restore (P1)	Not measured	Under 2 h	Incident management
Cost attribution coverage	0%	100% of consumption tagged	Billing telemetry; untagged consumption reported as a defect
Cost per certified product	Not measurable	Published monthly; trending down per unit of consumption	Attribution reporting
Analyst self-service rate	~15% (est.)	Over 70% of questions answered without engineering involvement	Request telemetry vs query telemetry
Time to onboard a new analyst	3–4 weeks	Under 1 day	Group membership to first successful query
Lineage coverage	0%	100% of certified products with end-to-end column lineage	Lineage system tables

SLO	Baseline	Target	How measured
Erasure SLA	Manual, unmeasured	100% executed within 14 days; physical removal within 21	Erasure control table
Deployment lead time	6–8 weeks	Under 2 days from merge to production	CI/CD telemetry
Changes authored by client engineers	0%	Over 80% for three consecutive months (handover gate)	Repository telemetry

End of document. Meridian Retail Group — Global Omnichannel Lakehouse · Solution design — executive brief · 27 September 2026 · Illustrative — fictional client; all figures illustrative.